

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Title

E-Commerce Order Management Database System

Introduction

The Software Requirement Specification (SRS) defines the functional and non-functional requirements of the E-Commerce Order Management Database System. It provides a clear understanding of the system's features, objectives, and expected performance.

Purpose

The purpose of this system is to develop a centralized database for managing customers, products, suppliers, orders, payments, shipments, and reviews efficiently.

Scope

The system supports:

- Customer Management
- Product Management
- Category Management
- Supplier Management
- Order Management
- Payment Management
- Shipment Management
- Review Management
- Report Generation

Overall Description

The E-Commerce Order Management Database System enables customers to purchase products online while administrators manage products, suppliers, orders, and payments. The system stores all business data in a centralized database for easy access and management.

Functional Requirements

- Customer Registration

- Customer Login
- Product Management
- Category Management
- Supplier Management
- Order Placement
- Order Details Management
- Payment Processing
- Shipment Tracking
- Customer Review Management
- Report Generation

Non-Functional Requirements

- Security
- Reliability
- High Performance
- Scalability
- Availability
- Data Integrity
- User-Friendly Interface
- Backup and Recovery

Use Cases

- Customer Registration
- Customer Login
- Browse Products
- Place Order
- Make Payment
- Track Shipment
- Submit Review
- Manage Products
- Generate Reports

Data Dictionary

Customer – Stores customer information such as Customer ID, Name, Email, Phone Number, and Address.

Category – Stores product category details.

Product – Stores product information such as Product ID, Name, Price, Stock Quantity, and Category.

Supplier – Stores supplier details including Supplier ID, Name, and Contact Information.

Order – Stores order details such as Order ID, Customer ID, Order Date, and Total Amount.

Order Details – Stores information about products included in each order.

Payment – Stores payment details such as Payment ID, Payment Method, Payment Date, and Payment Status.

Shipment – Stores shipment information including Shipment ID, Delivery Address, Shipment Date, and Delivery Status.

Review – Stores customer ratings and feedback for products.

System Features

- Secure User Authentication
- Product Search
- Inventory Management
- Order Processing
- Online Payment
- Shipment Tracking
- Customer Reviews
- Business Reports

System Constraints

- Internet connection is required.
- Only authorized users can access the system.
- Each customer must have a unique ID.
- Every order must be associated with a registered customer.
- Every payment must be linked to an order.

Future Enhancements

- Mobile Application Support
- AI-Based Product Recommendation
- Multiple Payment Options
- Email & SMS Notifications
- Sales Dashboard
- Advanced Analytics

Conclusion

The Software Requirement Specification provides a complete description of the E-Commerce Order Management Database System. It defines the system requirements, features, constraints, and future enhancements. This document serves as a guide for database design, implementation, testing, and maintenance.